

Multivariate Time Series Anomaly Detection in IIoT Using Spatial-temporal Dynamic Mask Diffusion Model

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Abstract—In recent years, multivariate time series anomaly detection has become an important research topic in the field of anomaly detection. In Industrial Internet of Things (IIoT) systems, the collected data may be affected by internal failures, external disturbances, or other adverse factors. In such cases, appropriate anomaly detection methods are required to ensure the stable operation of the system. However, existing methods based on reconstruction, prediction, or hybrid approaches often suffer performance degradation when anomalies are present in large amounts of training data, as these anomalies can negatively impact the training process. To address this challenge, we propose a dynamic masking strategy in both temporal and spatial dimensions. We develop a time series imputation framework based on a diffusion model that integrates Graph Neural Network (GNN) and Transformer architectures. This framework, termed Spatial-Temporal Dynamic Mask Diffusion for Anomaly Detection (STDMD-AD), incorporates a dynamic masking mechanism: temporally, reconstruction errors are used to mask data by randomly concealing values with higher errors; spatially, attention is applied to mask channels that are more likely to contain anomalies during training. Experiments on five real-world datasets demonstrate that the proposed method outperforms existing benchmarks and achieves state-of-the-art anomaly detection performance.

Index Terms—multivariate time series, anomaly detection, Industrial Internet of Things (IIoT), diffusion models, transformer.

I. INTRODUCTION

INDUSTRIAL Internet of Things (IIoT) [1]–[3] refers to the integration of internet technologies with the industrial sector by applying IoT technologies to manufacturing, energy, transportation, logistics, and other industrial fields to achieve interconnection of devices, systems, and factories. In the operation of large-scale systems related to this technology, the system heavily relies on the analysis of collected logs and other data. Internal failures or negative external impacts may occur, leading to anomalies that make it difficult for the system or program to achieve the expected performance in practical applications. This situation often arises in network

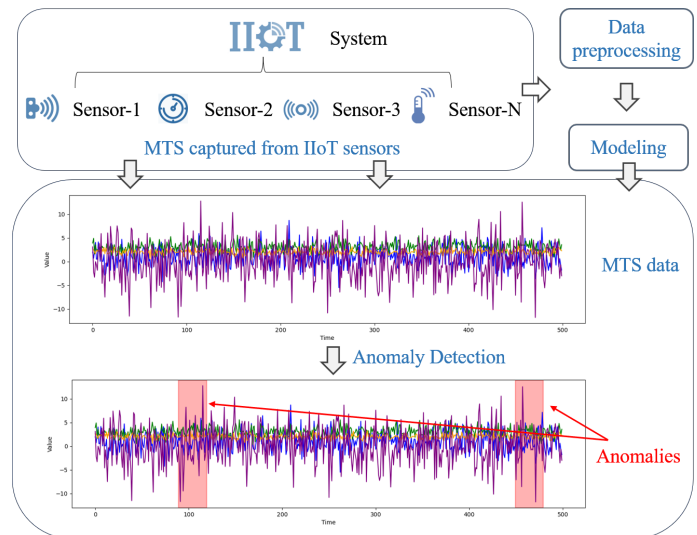


Fig. 1. The framework for anomaly detection in IIoT is shown below in the line graph of detecting anomalies (highlighted in red) from multidimensional time series data obtained from sensors.

security traffic monitoring [4], cyberattack [5], infrastructure monitoring [6], and fault diagnosis and system defects in the industrial field [7]. To address these issues and detect anomalies in the collected data, anomaly detection technology has emerged.

Anomaly detection techniques have applications in many fields and can be used to detect anomalies in data types such as images, graphs, and time series. The goal of these methods is to identify samples in the data that deviate significantly from other observations [8]. Multivariate time series anomaly detection is used to identify anomalies, abnormal time periods, or sudden events in multivariate time series data. In IIoT, the anomaly detection framework is illustrated in Fig. 1. It primarily involves monitoring data through multiple sensors and performing anomaly detection on the acquired multidimensional data to determine when anomalies occur. With the rapid development of industrial intelligence, this method has been proven effective in enabling the timely detection and collection of problems in data, thereby avoiding excessive losses. By leveraging advanced algorithms, this approach enhances a system's ability to respond to unexpected events, thereby improving operational efficiency. Furthermore, the integration

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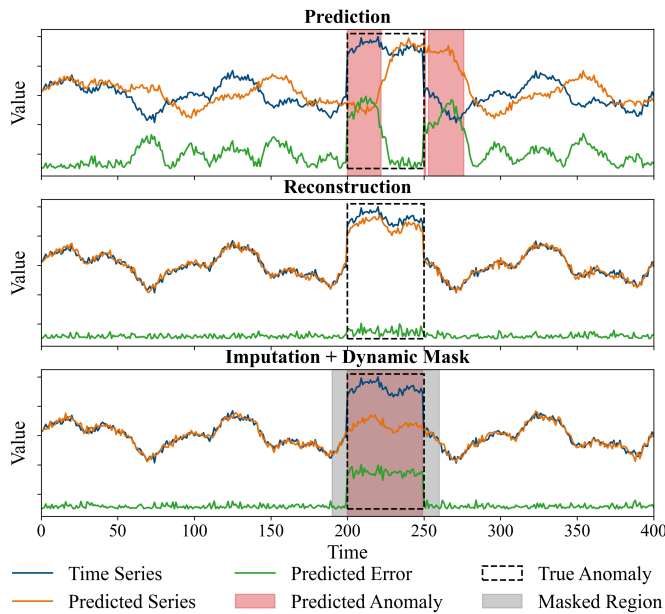


Fig. 2. A motivating example comparing failure modes of existing methods and our approach. Top two panels show prediction/reconstruction methods failing due to stochasticity or overfitting anomalies. Bottom panel demonstrates our strategy effectively ignoring anomalies, restoring normal patterns, and creating a detectable error gap.

of anomaly detection in IIoT systems facilitates predictive maintenance, allowing proactive measures to be taken before issues escalate into significant failures.

Currently, most research focuses on unsupervised methods, mainly because anomalies usually account for only a small proportion of the collected data and are difficult to label in large quantities [7]. Traditional approaches include clustering-based and density estimation-based methods. However, these methods rely heavily on manually crafted features and struggle to ensure model robustness when dealing with increasingly complex data. In recent years, with the rapid development of deep learning, anomaly detection methods that integrate deep learning have achieved significant performance improvements and demonstrated strong practical effectiveness [9]. Those methods can generally be categorized into three groups: reconstruction-based, prediction-based [10], and hybrid approaches [11]. Reconstruction-based methods determine anomalies by evaluating reconstruction errors, while prediction-based methods rely on deviations between predicted and actual data. Although these methods have achieved promising results, two major challenges remain in this task.

- The high-dimensional nature of Industrial Internet of Things (IIoT) data, accompanied by its intricate internal dependencies, necessitates the development of models with superior representational capacity. [12] Typical IIoT systems employ a multitude of sensors that collectively generate voluminous, uncertain, and highly correlated multivariate time-series data. Conventional reconstruction and prediction-based anomaly detection algorithms often exhibit limited efficacy in such challenging settings. A

motivating example illustrating these limitations is presented in Fig. 2. Prediction-based methods often struggle with the intrinsic stochasticity of future observations, leading to deviations in normal regions or insensitivity to actual anomalies. Similarly, the Reconstruction panel demonstrates a generalization trap where the model learns to reconstruct the anomalous patterns alongside the normal data. Consequently, the reconstruction error remains low even during anomalies, causing missed detections.

- The uneven distribution of anomalies within real-world IIoT data streams gives rise to the issue of anomaly concentration, which substantially complicates both data preprocessing and model training [3]. Owing to the impracticality of manual annotation for large-scale multivariate time series, anomaly detection models frequently depend on unsupervised learning paradigms. In such settings, the inadvertent inclusion of anomalous instances in the training set—such as contiguous fault sequences misleadingly treated as normal—can severely disrupt the learning dynamics, causing the model to assimilate abnormal patterns into its representation of normal behavior. For instance, reconstruction-based methods are prone to regenerating anomalous patterns—especially when such anomalies recur frequently during training. In contrast, as illustrated in the Imputation + Dynamic Mask panel of Fig. 2, this method proactively masks the suspected region. By forcing the model to infer the missing segment based solely on the surrounding spatial-temporal context, it recovers the underlying normal pattern, effectively rejecting the anomalous inputs. This creates a distinct discrepancy between the predicted series and the true anomaly, thereby avoiding the generalization trap and significantly enhancing detection sensitivity.

To directly address the challenges, we propose the Spatio-Temporal Diffusion Model for Detection (STDMD). Our design makes explicit, targeted improvements to the diffusion model framework to mitigate the specific issues of anomaly reconstruction and model contamination. Mitigating Challenge 1 (Complex Dependencies): Instead of prediction or reconstruction-based approaches, which struggle with complex dependencies and often regenerate anomalies, our model is built upon a time series imputation framework (Conditional score-based diffusion models, CSDI [13]). This foundation provides a key advantage: by learning to accurately impute masked normal values based on their highly correlated spatio-temporal context, the model learns a robust representation of the system's healthy dynamics. To further enhance its capacity to capture intricate dependencies, we integrate a Masked Graph Network and a Transformer as the feature extraction backbone. This hybrid design explicitly models the complex spatial relationships between sensors (via the Graph Network) and long-range temporal patterns (via the Transformer), enabling the model to discern subtle deviations that constitute anomalies, rather than reconstructing them.

Mitigating Challenge 2 (Anomaly Concentration): To prevent the model from assimilating anomalies present in the unsupervised training set, we introduce a novel Dual Dynamic

Masking Mechanism. The Temporal Dynamic Mask applied to the input of the diffusion model randomly occludes contiguous segments of the time series. This prevents the model from over-relying on any specific time period that might be contaminated by a concentrated anomaly, forcing it to learn a generalized representation of normality from the unmasked context. The Spatial Dynamic Mask Attention mechanism operates on the feature level. It adaptively attenuates the contribution of sensor channels (features) that exhibit large reconstruction errors during training. This dynamically reduces the influence of potentially anomalous channels, preventing them from polluting the model's latent representations.

Through this dual strategy, we proactively preprocess potentially anomalous regions in both the temporal and feature domains, thereby significantly mitigating their adverse effects on model training and enhancing overall robustness. The core architecture of our STDMD-AD is built on an existing conditional denoising diffusion model originally designed for time series imputation, and all our contributions represent targeted modifications, innovative component designs and systematic integrations rather than simple adoption of off-the-shelf models, with each design tailored to the unique challenges of multivariate time series anomaly detection. The contributions of this paper are as follows:

- We propose a novel anomaly detection framework, Spatial-temporal Dynamic Mask Diffusion Anomaly Detection (STDMD-AD), which incorporates time series interpolation and leverages temporal and spatial dynamic masks. The framework integrates Graph Neural Networks (GNN), Transformers, and Denoising Diffusion Models to achieve precise anomaly detection. Anomalies are identified by selectively masking regions with higher anomaly likelihood and gradually denoising them to recover the original signals.
- We introduce a temporal dynamic masking method that utilizes a Transformer-based encoder for data reconstruction. The input data is preprocessed by masking values with large reconstruction errors, which are indicative of potential anomalies, to hide these regions before model training. This approach alleviates the negative impact of high anomaly concentrations on the training process.
- We incorporate a dynamic mask attention mechanism into the spatial layer of the network to mask anomalous regions during training. This allows the model to focus on cleaner, more representative data, thereby improving its learning efficiency and detection performance.

The remainder of this paper is structured as follows: The related work is presented in the Section II. The section III provides a detailed introduction to the proposed STDMD-AD framework and each module, while the section IV includes experimental results, comparisons, and analysis of ablation experiments. Finally, Section V summarizes the work.

II. RELATED WORK

In this section, we briefly introduce the methods of multivariate time series anomaly detection based on prediction, reconstruction and diffusion models, including both traditional

and deep learning approaches, with a specific focus on studies modeling correlations between multi-sensor data.

A. Multivariate Time Series Anomaly Detection

Multivariate time series anomaly detection has been a significant focus for researchers and the academic community. In the realm of deep learning, methods in this field can be primarily categorized into three types [7]: (1) Prediction-based methods, which utilize models for forecasting time series data, such as Long Short-Term Memory (LSTM) [14] and Gated Recurrent Units (GRU) [15], to predict future values and compare them with actual values to detect anomalies. (2) Reconstruction-based methods, which employ generative models like Generative Adversarial Networks (GANs) [16]–[19] and Variational Autoencoders (VAEs) [20], [21] to train and reconstruct data, creating pseudo-samples that are then compared to real samples to identify anomalies. (3) Hybrid methods, which typically combine the above two approaches, using a fusion of results from various methods to obtain the final outcome.

In recent literature, prediction and reconstruction-based methods have become popular for their effectiveness in anomaly detection, with many typical works focusing on modeling multi-sensor correlations to enhance detection performance. For example, OmniAnomaly [22] combines GRU and VAE to form a stochastic recurrent neural network model, learning the normal patterns of multivariate time series and using reconstruction probability for anomaly detection. GDN [23] constructs graph structures based on multi-sensor physical connections and proposes a cross-network meta-learning algorithm to aggregate inter-sensor information for few-shot anomaly detection. MTAD-GAT [24] leverages Graph Attention Networks to learn adaptive spatial correlations between sensors and capture temporal trends, enhancing detection performance through a joint prediction and reconstruction model. TranAD [25] incorporates an attention-based sequence encoder into the Transformer model to implicitly capture multi-sensor feature correlations via embedding fusion, combining adversarial training to improve model accuracy. In addition, other graph-based methods [26]–[29] have also been utilized by researchers for modeling inter-sensor dependencies in anomaly detection and have achieved promising results. However, these methods either rely on prior physical knowledge for static graph construction, ignore the dynamic coupling of spatial and temporal dependencies between sensors, or suffer from interference of anomalous sensor segments when learning valid correlations.

The method proposed in this paper differs from the above categories. It primarily utilizes time series interpolation and applies **dynamic masking in both temporal and spatial dimensions** to model multi-sensor spatiotemporal correlations adaptively, which mitigates the interference of anomalous data on dependency learning and thus enhance anomaly detection performance.

B. Diffusion model

In recent years, the emergence of denoising diffusion models [30] has brought about significant changes in the field

of image generation. Their powerful generative capabilities have made them perform excellently in downstream tasks such as text generation, audio generation, and image anomaly detection. In the field of image generation, Wolleb et al. [31] proposed a weakly supervised anomaly detection method based on the denoising diffusion implicit model, which is used to generate abnormal maps of regions affected by disease in patients. Wyatt et al. incorporated multi-scale (multi-octave) simplex noise into the diffusion model, replacing the original Gaussian noise, ultimately introducing AnoDDPM [32] for the detection of anomalies in medical images.

In the field of time series data generation, Kong et al. [33] introduced a universal diffusion probabilistic model called DiffWave for original audio synthesis, which can generate high-quality audio signals for conditional and unconditional waveform generation tasks. Currently, there are several denoising diffusion model-based methods in time series research that also explore multi-sensor correlation modeling for imputation and anomaly detection. For example, Alcaraz et al. [34] combined the diffusion model with structured state space models, proposing a new time series imputation method for multi-dimensional sensor data. Xiao et al. [35] extracted normal observations using a density ratio-based selection strategy and then used a conditional weighted incremental diffusion model to estimate missing values of multi-sensor time series. Chen et al. [36] designed a raster mask strategy to create missing values in the data and used ImTransformer with a spatial Transformer layer to capture inter-sensor correlations, training the diffusion model for time series imputation and subsequent anomaly detection tasks. Zeng et al. [37] proposed a wireless anomaly detection algorithm AE-DDPMs for wireless communication security, based on an improved denoising diffusion probabilistic model to model correlations among wireless sensor data. These methods have achieved high performance on public datasets but adopt fixed modeling or masking strategies for multi-sensor dependencies, lacking dynamic adaptation to anomalous data segments.

To the best of our knowledge, while our method is not the first to use diffusion models for multivariate time series anomaly detection, our approach to data masking is more comprehensive. We perform **dynamic masking operations in both temporal and spatial dimensions** to adaptively focus on clean and representative multi-sensor spatiotemporal correlations, which makes the modeling of inter-sensor dependencies more robust to anomalous data compared with existing diffusion-based methods.

III. METHODOLOGY

In this section, we first briefly discuss the issues of multivariate time series anomaly detection. Then, we provide a detailed explanation of the space-temporal dynamic masking mechanism. After that, we elaborate the Space-temporal dynamic mask Diffusion for Anomaly Detection (SDMD-AD) framework and its process.

A. Problem Overview

Given a multivariate time series dataset $S = s_1, s_2, \dots, s_N$, where each s_i is a D -dimensional vector representing the val-

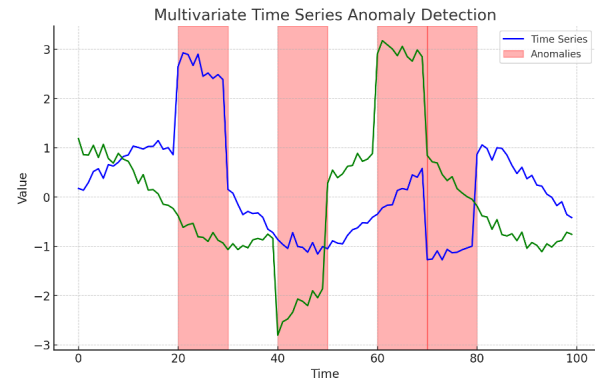


Fig. 3. Line chart of an example of multivariate time series anomaly detection task. The multidimensional time series is represented by blue and green lines, and all anomalies are represented by red shaded areas.

ues of D features. Our goal is to detect outliers in the dataset, i.e., samples that are far from the majority of observations in the feature space. These outliers may represent underlying issues or anomalous events.

To solve this problem, we can use an anomaly detection model M that maps each sample s_i to an anomaly score ad_i , indicating the level of anomaly of sample s_i . Typically, we expect higher ad_i to indicate a higher likelihood of sample s_i being an outlier. As shown in the Fig. 3, our ultimate goal is to identify which points or segments in the data are considered anomalies, thus completing the anomaly detection task.

In our paper, We frame anomaly detection as a self-supervised imputation task, where a binary mask matrix $M \in \{0, 1\}^{T \times D}$ is applied to the original series S to generate the masked input $S_{masked} = S \odot M$. The model is then trained to reconstruct the masked values from the observed context. Unlike standard random masking, our mask M is dynamically constructed by integrating two strategies detailed in Section III-B: the Temporal Dynamic Mask, which conceals time steps with high reconstruction errors, and the Spatial Dynamic Mask, which targets channels exhibiting anomalous patterns. This targeted masking strategy forces the model to recover likely anomalies rather than random missing values, thereby learning more robust representations of normal dynamics.

B. Spatial-temporal dynamic mask

Due to the presence of unlabeled outliers or segments in the training dataset, in order to minimize their negative impact on model training, this paper proposes two mask strategies from both temporal and spatial dimensions: temporal dynamic mask and Spatial dynamic mask.

1) *Temporal dynamic mask*: In terms of the time dimension masking strategy, we first utilize a Transformer encoder to reconstruct the original data and compute the reconstruction error with the Mean Squared Error (MSE) loss between the reconstructed and original data. A threshold is then derived from the reconstruction error, and points with values exceeding this threshold are considered more likely to be anomalous. Finally, a masking ratio is set to randomly conceal data with larger reconstruction errors. This strategy serves as a preprocessing step for the input data in the temporal dimension.

Specifically, during the training phase, the raw data are fed into the encoder network, which outputs a pseudo-sample of the same dimensionality. To robustly identify potential anomalies without relying on fixed error thresholds, we employ a *confidence-based ranking mechanism*. First, we compute the element-wise reconstruction error (MSE) and transform it into a confidence score S , formulated as:

$$S = \frac{1}{1 + \mathcal{L}_{MSE}(x, \hat{x})} \quad (1)$$

where a higher reconstruction error corresponds to a lower confidence score. Subsequently, we sort these scores and derive a dynamic threshold based on the α -quantile. Points falling into the bottom 5% of confidence scores are flagged as potential anomalies and selected as candidates for masking.

Multivariate anomalies in IIoT systems typically manifest as persistent shifts in system states rather than instantaneous, isolated outliers. To capture this temporal continuity, we extend the masking mechanism to adjacent time points that exhibit similar spatial patterns. Let $\mathbf{x}_t \in \mathbb{R}^D$ denote the feature vector of a candidate anomaly at time t , where D represents the number of sensor channels. To delineate the effective range of this dependency, we introduce a temporal search radius w . Consequently, the identification of potential neighbors is constrained to the window $[t-w, t+w]$ centered at the anomaly candidate. For each neighboring time step t' within this local window, we quantify the structural similarity between the candidate $\mathbf{x}_t$ and the neighbor $\mathbf{x}_{t'}$ using the Pearson correlation coefficient. A high correlation implies that the system state at t' mirrors the anomalous profile at t , thereby necessitating its inclusion in the mask. The Pearson correlation coefficient [38] is calculated across the feature dimension as follows:

$$r_{t,t'} = \frac{\sum_{d=1}^D (x_{t,d} - \bar{x}_t)(x_{t',d} - \bar{x}_{t'})}{\sqrt{\sum_{d=1}^D (x_{t,d} - \bar{x}_t)^2} \sqrt{\sum_{d=1}^D (x_{t',d} - \bar{x}_{t'})^2}} \quad (2)$$

where $x_{t,d}$ represents the value of the d -th channel at time t , and $\bar{x}_t = \frac{1}{D} \sum_{d=1}^D x_{t,d}$ denotes the spatial average across all channels at that specific timestamp. This metric robustly evaluates the similarity of the multivariate distribution shapes between the two time steps, ensuring that the masked region covers the entire duration of the anomalous event.

Finally, simply masking all points identified by the error and correlation criteria might lead to excessive information loss, making model convergence difficult. To balance this, we introduce a stochastic sampling strategy controlled by a masking ratio parameter γ . From the expanded candidate set obtained in the previous steps, we randomly select γ proportion of points to generate the final binary mask. This stochasticity acts as a regularization technique, preventing the model from overfitting to specific masking patterns.

The T-Encoder network used for reconstruction consists of one encoder layer from our Spatial-Temporal Transformer and a simple decoder layer. Although the dynamic masking strategy may occasionally conceal non-anomalous values, it effectively reduces the negative impact of anomalous data on model training and enhances the model's robustness and generalization, thereby improving anomaly detection performance.

2) *Spatial dynamic mask*: For the spatial dimension masking strategy, we develop a dynamic mask attention mechanism applied to the channels. Since certain channels may contain a higher proportion or density of anomalous data, this attention mechanism is integrated into the diffusion model. The procedure is as follows: first, each channel is treated as a node in a graph; second, the attention feature of each node is computed; and finally, the overall mean and standard deviation of these features are calculated. Channels with feature values lower the sum of the mean and one standard deviation are then masked.

Given the input feature map $\mathbf{X} \in \mathbb{R}^{C \times D}$, where C is the number of channels and D is the feature dimension. Unlike standard global attention which computes a $C \times C$ dependency map, we focus the query solely on the target channel to evaluate its relationship with all other channels. We first generate the query vector for the target channel ($\mathbf{q}_{target}$), and the key ($\mathbf{K}$) and value ($\mathbf{V}$) matrices for all channels through linear projections:

$$\mathbf{q}_{target} = \mathbf{x}_{target} W_Q, \quad \mathbf{K} = \mathbf{X} W_K, \quad \mathbf{V} = \mathbf{X} W_V \quad (3)$$

where W_Q, W_K, W_V are learnable weight matrices. We then compute the raw attention scores vector $\mathbf{z} \in \mathbb{R}^C$ by performing the dot product between the target query and all keys:

$$\mathbf{z} = \frac{\mathbf{q}_{target} \mathbf{K}^T}{\sqrt{d_k}} \quad (4)$$

where d_k is the scaling factor. Each element z_i in $\mathbf{z}$ represents the relevance of the i -th channel to the target channel.

To mask channels that contain anomalies or provide irrelevant information, we introduce a dynamic thresholding strategy. We calculate a threshold θ based on the statistical distribution of the attention scores $\mathbf{z}$:

$$\theta = \text{mean}(\mathbf{z}) + \text{std}(\mathbf{z}) = \frac{1}{C} \sum_{i=1}^C z_i + \sqrt{\frac{1}{C} \sum_{i=1}^C (z_i - \bar{z})^2} \quad (5)$$

Channels with scores below this threshold are deemed uninformative. Consequently, we define a set of valid channel indices Set to serve as the effective receptive field for the target channel:

$$Set = \{i \mid z_i \geq \theta, \quad i \in 1, \dots, C\} \quad (6)$$

We then calculate the normalized attention weights a_i only for the channels within the valid set using the Softmax function:

$$a_i = \frac{\exp(z_i)}{\sum_{j \in Set} \exp(z_j)}, \quad \forall i \in Set. \quad (7)$$

Finally, we reconstruct the feature of the target channel by aggregating the values from the valid set: $\mathbf{v}'_{target} = \sum_{i \in Set} a_i \mathbf{v}_i$. To integrate this reconstructed feature back into the multivariate time series without altering the non-target channels, we employ a channel-wise concatenation operation followed by a linear projection W_O :

$$\mathbf{Y} = W_O \cdot \text{Concat}(\mathbf{v}'_{target}, \mathbf{V}_{others}) \quad (8)$$

where $\mathbf{V}_{others}$ denotes the projected features of the non-target channels, and $\text{Concat}(\cdot)$ splices the reconstructed target feature

with the original features of the remaining channels. This design ensures that the model selectively repairs the anomaly using only high-confidence spatial contexts.

C. STDMD-AD

1) **Overview:** The overall architecture of STDMD-AD, as shown in Fig. 4. It can be seen that our framework is mainly divided into three parts: training of the diffusion model based on the spatial-temporal transformer, offline validation of the model and the main framework of the spatial-temporal transformer, including the structure of its two main feature extraction layers: the spatial layer and the temporal layer. Moreover, Table I presents a comparison of hyperparameters between STDMD-AD and the baseline models. It can be observed that both STDMD-AD and the baselines utilize the Spatial-temporal Transformer and share the same settings for several hyperparameters such as the number of feature extraction layers, batch size, time embedding dimension, feature embedding dimension, diffusion steps, and window size. However, a notable difference lies in the adoption of dynamic masks: STDMD-AD incorporates both temporal dynamic mask and spatial dynamic mask, while the baseline models do not employ these dynamic mask mechanisms.

TABLE I
COMPARISON OF COMPONENTS AND HYPERPARAMETERS BETWEEN
STDMD-AD AND BASELINES

Hyper parameters \ Model	STDMD-AD	Baselines
Spatial-temporal Transformer	✓	✓
Temporal dynamic mask	✓	x
Spatial dynamic mask	✓	x
Feature extraction layers	4	4
Batch size	32	32
Time embedding dim	128	128
Feature embedding dim	16	16
Diffusion steps	50	50
Window size	100	100

2) **Denoising Diffusion Model:** The Denoising Diffusion Implicit Models (DDIMs) is a more efficient iterative implicit probabilistic model, building upon the Denoising Diffusion Probabilistic Models (DDPMs) while maintaining the same training process. DDIMs enables the generation of high-quality samples with fewer iteration steps, significantly speeding up the sampling process. The sample generation procedure in DDIMs primarily involves two key stages: the forward diffusion process and the reverse diffusion process.

Forward noise addition process is consistent with that in DDPMs, continuously adding Gaussian noise to data, which can be represented as:

$$q(s_t | s_{t-1}) := \mathcal{N}(s_t; \sqrt{1 - \beta_t} s_{t-1}, \beta_t \mathbf{I}) \quad (9)$$

where $\{\beta_t\}_{t=1}^T$ is the variance of the Gaussian noise. Subsequently, according to the definition in the paper: $a_t := 1 - \beta_t$ and $\bar{a}_t := \prod_{s=1}^t a_s$, the formula for s_t can be obtained as:

$$s_t = \sqrt{\bar{\alpha}_t} s_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \text{with } \epsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (10)$$

Then, based on equation (10), the forward noise addition formula can be derived as follows:

$$s_{t+1} = s_t + \sqrt{\bar{\alpha}_{t+1}} \left(\sqrt{\frac{1}{\bar{\alpha}_t}} - \sqrt{\frac{1}{\bar{\alpha}_{t+1}}} \right) s_t + \sqrt{\bar{\alpha}_{t+1}} \left(\sqrt{\frac{1}{\bar{\alpha}_{t+1}} - 1} - \sqrt{\frac{1}{\bar{\alpha}_t} - 1} \right) \epsilon_\theta(s_t, t). \quad (11)$$

Reverse diffusion processes differs from DDPMs, where the reverse process is still a Markov chain that depends on the Bayesian formula. The time step must be the same as the forward process to generate samples step by step. However, in DDIMs, the sampling process does not depend on the Bayesian formula. The defined generation process is:

$$p_\theta^{(t)}(s_{t-1} | s_t) = \begin{cases} \mathcal{N}(f_\theta^{(1)}(s_1), \sigma_1^2 \mathbf{I}) & \text{if } t = 1; \\ q_\sigma(s_{t-1} | s_t, f_\theta^{(t)}(s_t)) & \text{otherwise.} \end{cases} \quad (12)$$

Based on the derivation in the DDIMs paper, the formula for st can be obtained as:

$$s_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \left(\frac{s_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta^{(t)}(s_t)}{\sqrt{\bar{\alpha}_t}} \right) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2 \epsilon_\theta^{(t)}(x_t)} + \sigma_t \epsilon_t \quad (13)$$

where $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$ is standard Gaussian noise, $\sigma_t \epsilon_t$ represents random noise, and when $\sigma_t = \sqrt{(1 - \bar{a}_{t-1})/(1 - \bar{a}_t)} \sqrt{1 - \bar{a}_t/\bar{a}_{t-1}}$ for all t , the forward process becomes a Markov process and the reverse process becomes the process of DDPMs; when $\sigma_t = 0$, the reverse process becomes the deterministic sampling process in DDIMs.

For the optimization objective of this time series imputation task, we also use the MSE loss for training, following the loss calculation approach from the CSDI. During training, the observed values s_0^{co} are used as conditional inputs, while s_0^{ta} represents the imputation target, i.e., the masked missing values. The known noise ground truth is ϵ , and $\epsilon_\theta(\sqrt{\bar{\alpha}_t} s_0^{ta} + \sqrt{1 - \bar{\alpha}_t} \epsilon, t | s_0^{co})$ is the noise predicted by the model. The MSE loss is then computed between these predicted and true values to train the model:

$$\mathcal{L} := \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} s_0^{ta} + \sqrt{1 - \bar{\alpha}_t} \epsilon, t | s_0^{co})\|_2^2, \quad \text{with } \epsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (14)$$

3) **Spatial-temporal Transformer Backbone:** To effectively model the complex dependencies in multivariate time series, we employ a Spatial-Temporal Transformer as the denoising backbone, replacing the standard U-Net architecture typically used in diffusion models. As illustrated in Fig. 4(a), the framework processes data through a sequential pipeline of embedding injection, spatio-temporal feature extraction, and signal reconstruction.

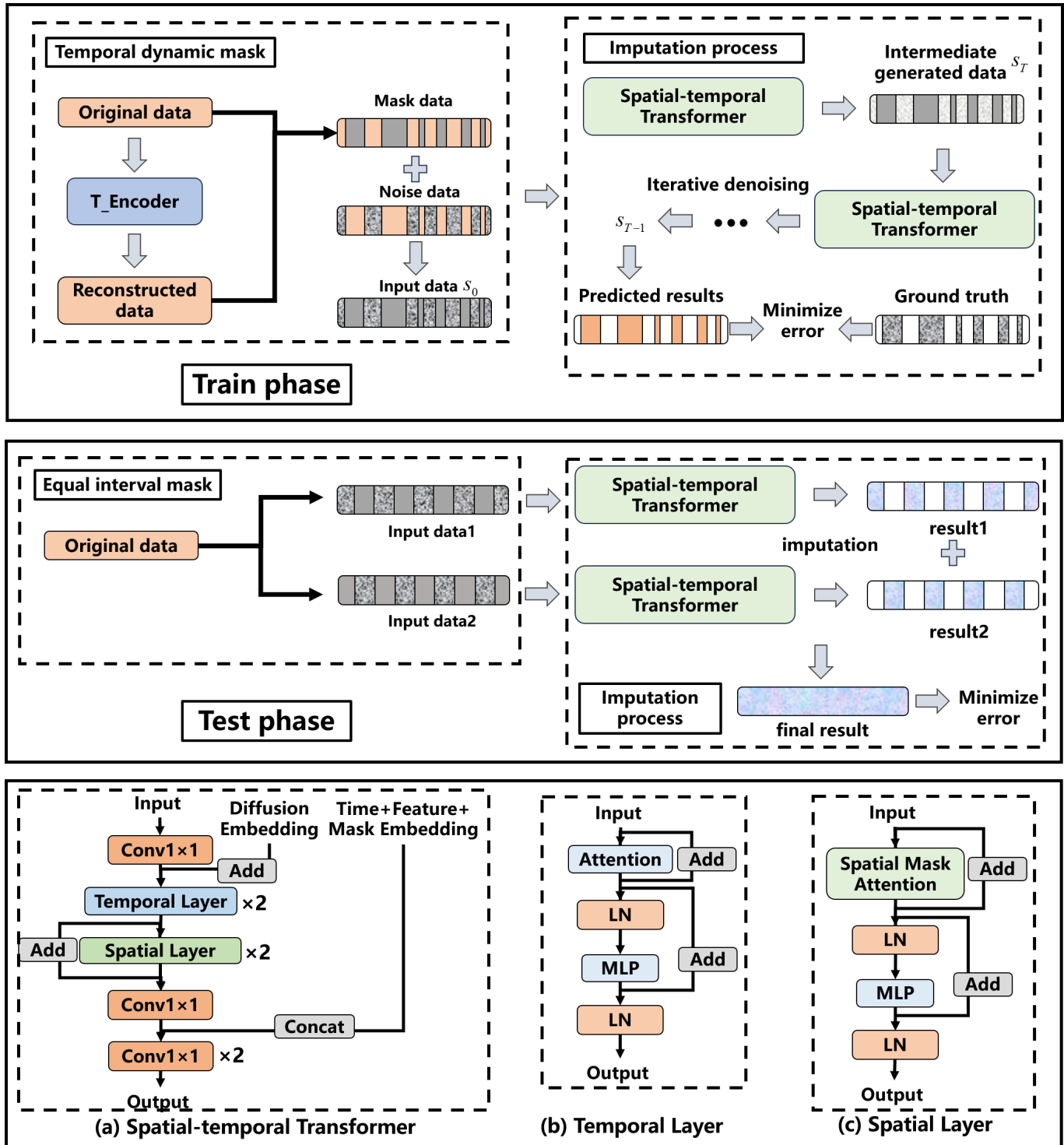


Fig. 4. The overall framework diagram of the spatial-temporal dynamic mask diffusion for anomaly detection. It showcases three parts: the training and testing process, and the general structure of the network model.

The network first accepts the masked noisy input projected into a high-dimensional space. To preserve context during the iterative denoising process, we fuse the input with auxiliary embeddings. Specifically, we introduce a Diffusion Embedding to indicate the current noise level and a joint spatio-temporal embedding that encodes temporal positions and channel identifiers.

These embeddings are then processed by the core extraction unit, which consists of stacked Temporal Layers and Spatial Layers. The Temporal Layers utilize self-attention mechanisms to capture long-range dependencies along the time axis. Subsequently, the Spatial Layers model inter-variable correlations at each time step. Notably, our proposed *Spatial Dynamic Mask Attention* is integrated within these spatial layers to explicitly suppress anomalous channels.

To facilitate gradient propagation and feature integration, we employ a global residual connection that fuses the input features with the processed representations. Finally, a 1×1 convolutional projection layer maps the hidden features back to the original data dimensions, yielding the predicted noise or reconstructed signal.

4) **Training and Testing Process of STDMD-AD:** The training process of our method is illustrated in the top block of Fig.4. First, data preprocessing is performed using the Temporal Dynamic Mask. The input is fed into a Transformer encoder for reconstruction, and masking positions are determined based on the MSE loss between the reconstructed and original data, along with correlations between adjacent points. Noise is then added to the unmasked data, and combined with the masked data to form the model input. Gaussian noise is applied to train the spatial-temporal Transformer, which gradually learns to denoise and predict the masked values. Finally, the loss is computed between the predicted and true values to optimize the model in inferring the masked normal data.

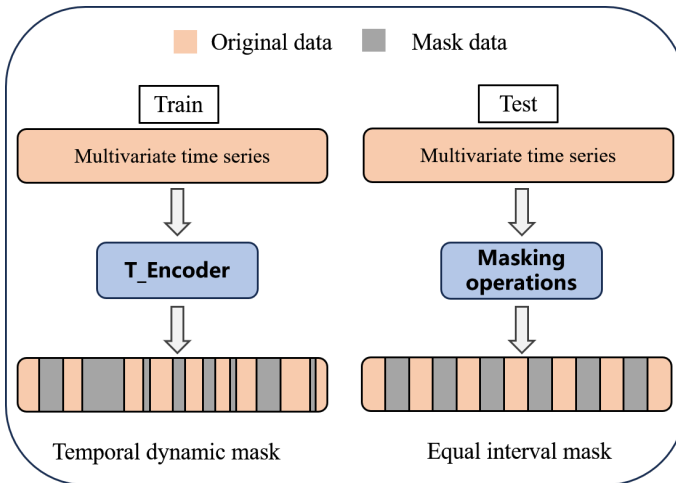


Fig. 5. Explanation of Masking Strategy in Training and Testing Phases: The left side shows the result processed with temporal dynamic mask in the training phase, while the right side shows the result processed with equal interval mask in the testing phase. The T-Encoder consists of one encoder layer from our Spatial-Temporal Transformer and a simple decoder layer.

As illustrated in Fig. 5, distinct masking strategies are em-

ployed for the training and testing phases. While the dynamic mask is crucial for learning robust features during training, we adopt an equal interval mask for the testing phase, as shown in the right panel of Fig. 5. This approach avoids the computational overhead associated with the dynamic mechanism, significantly reducing inference complexity.

The testing process of the framework is shown in the middle block of Fig. 4. Specifically, the data is first processed with equal interval mask, generating two inputs with identical masking intervals. These inputs are then independently fed into the model for time series interpolation, allowing it to capture the spatial and temporal correlations in the data and accurately reconstruct the masked values. The two outputs are subsequently concatenated to form the complete prediction, thereby improving reconstruction quality. Finally, an anomaly score is computed based on the difference between the reconstructed output and the ground truth. By applying the computed threshold to these scores, the framework determines whether each data point is anomalous, thereby enabling effective detection of abnormal events.

IV. EXPERIMENTS AND RESULTS

A. Datasets

We collect five real-world datasets to test our method. The following is a description of the five experimental datasets [3]:

Mars Science Laboratory (MSL) and Soil Moisture Active Passive (SMAP) [39]: The Mars Science Laboratory and Soil Moisture Active Passive satellites are public datasets provided by NASA. The Mars Science Laboratory dataset consists of 55 dimensions, while the Soil Moisture Active Passive satellite dataset consists of 25 dimensions. These datasets include telemetry anomaly data reported by the spacecraft detection system, known as incident unexpected anomalies (ISA).

Server Machine Dataset (SMD) [22]: The SMD dataset was collected at a large Internet company over a five-week period. It consists of 38-dimensional multivariate time series data.

Pool Server Metrics (PSM) [5]: PSM is collected from multiple application server nodes within eBay. It contains data from 26 dimensions.

Safe Water Treatment (SWaT) [6]: The SWaT dataset includes data from 51 sensors that are continuously operating in critical infrastructure systems, particularly water treatment facilities.

B. Experimental setup

The STDMD-AD framework is implemented based on PyTorch version 3.8 and is trained and tested on an NVIDIA Titan Xp device. The batch size is set to 32, with each data sample having a length of 100. A random masking ratio of 0.7 is applied, and the diffusion model's time step is set to 50. During training, a random search method is used to find the optimal combination of hyperparameters, while other comparison models follow the settings specified in their respective original papers.

C. Evaluation Metrics

Anomaly detection is a critical task in various domains, requiring robust methods to identify outliers or abnormal patterns within datasets. Evaluating the performance of anomaly detection models involves specific metrics tailored to assess their effectiveness in identifying anomalies. The primary evaluation metrics used in anomaly detection include Precision, Recall, and F1-Score.

Precision measures the proportion of true anomalies correctly identified by the model out of all the instances it classified as anomalies. It is defined as:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (15)$$

A high precision indicates that the model has a low false positive rate, meaning it does not incorrectly label normal data as anomalous frequently.

Recall, also known as Sensitivity or True Positive Rate, measures the proportion of actual anomalies that are correctly identified by the model. It is defined as:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (16)$$

A high recall indicates that the model successfully identifies most of the actual anomalies present in the dataset, minimizing the false negatives.

F1-Score is the harmonic mean of Precision and Recall, providing a single metric that balances both aspects. It is particularly useful when dealing with imbalanced datasets where the number of normal instances far outweighs the anomalies. The F1-Score is defined as:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (17)$$

The F1-Score ranges from 0 to 1, where a higher value indicates a better balance between precision and recall.

D. Ablation experiment

In this section, we conduct an ablation study to systematically investigate the impact of various components and hyperparameters on the performance of our proposed model. By isolating and modifying specific aspects of the model, we aim to understand their contributions and effectiveness in the context of anomaly detection in time series data.

1) *Model components*: The results of the ablation studies in Table II indicate that our STDMD-AD model consistently performs well across all five datasets. Specifically, the temporal and spatial dynamic masking strategies are critical for enhancing model performance. The removal of the temporal dynamic mask resulted in a significant decrease in F1 scores across all datasets. In the MSL, SMAP, PSM, SMD, and SWaT datasets, the performance dropped by 5.20, 4.52, 5.45, 5.12, and 7.97 percentage points, respectively. The findings demonstrate the key role of the temporal dynamic mask in improving the accuracy of time series anomaly detection. Similarly, removing the spatial dynamic mask also led to substantial performance degradation, particularly in the MSL, SMAP, and PSM datasets, with decreases of 6.44, 4.53, and

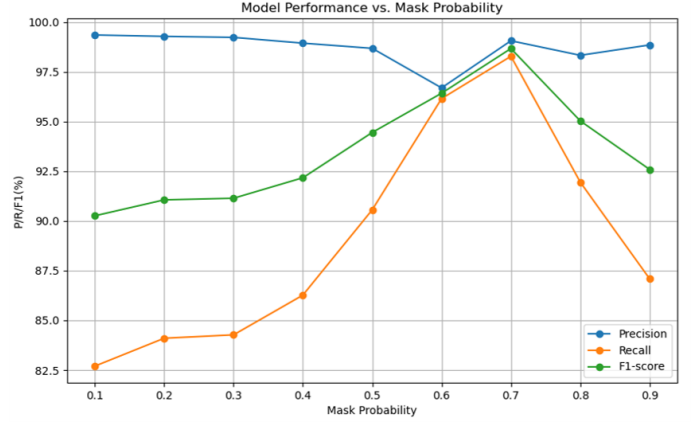


Fig. 6. The model's performance parameters P, R, and F1 as the mask probability changes on the dataset.

4.14 percentage points, respectively. These results validate the importance of spatial masking in capturing multivariate dependencies and processing anomalous features across channels. Overall, the ablation results confirm the rationality and effectiveness of our model design and provide a solid basis for further optimization.

Moreover, in terms of efficiency, the dynamic masking strategy only increases the training time per epoch by approximately 14.2% to 22.7% compared to static masking, primarily due to the iterative reconstruction process and adaptive threshold computation. However, even with dynamic masking, the maximum total training time across all datasets remains within 120 minutes. Furthermore, during the inference phase, we employ fixed-interval masking (static masking) for data preprocessing to ensure stable and consistent anomaly detection results. Consequently, the dynamic masking strategy has negligible impact on inference efficiency.

2) *Masking probability*: In our experiments, we conduct an ablation study to evaluate the impact of varying mask probabilities on the performance of our STDMD-AD model. Mask probability refers to the fraction of data points that are randomly masked during training. We test mask probabilities ranging from 0.1 to 0.9 and analyzed the results in terms of precision, recall, and F1-score across PSM datasets. It can be seen from the Fig. 6 that as the mask probability increases, the performance of the model continues to improve until the mask probability reaches 0.7, at which point the performance begins to decline. This is because increasing the mask probability obscures more anomalous information in the data, and a higher mask probability may cause the model to lose more important information during training, thereby affecting its ability to recognize true positives.

3) *Diffusion steps*: The sensitivity experiment and the results in Fig. 7 reveal significant differences in the model performance of various indicators as the diffusion step size increases from 10 to 80. Initially, from 10 to 50 steps, the model showed a significant improvement in F1 and Recall. However, as the diffusion steps further increased, the performance gains became less noticeable, and there might even be

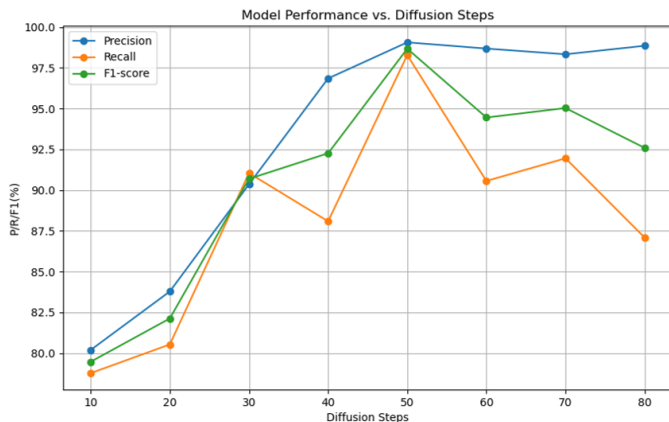


Fig. 7. The model's performance parameters P, R, and F1 as the diffusion steps changes on the dataset.

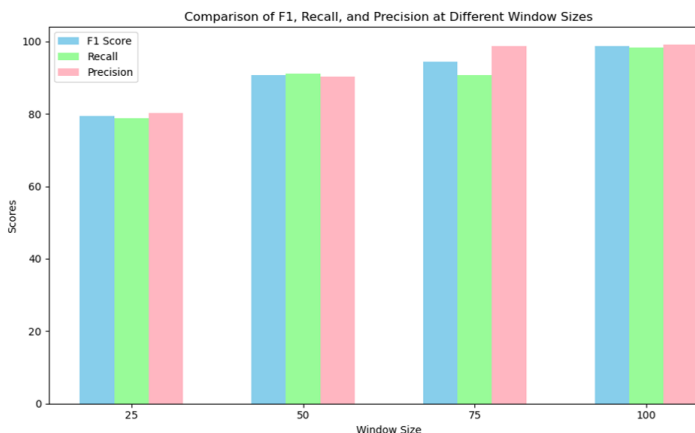


Fig. 8. The model's performance parameters P, R, and F1 as the window sizes changes on the dataset.

a slight decline after 50 steps. Specifically, the model showed an average precision increase of about 18% between 10 and 50 steps. Beyond 50 steps, while the model maintained a stable performance level, the rate of precision improvement slowed down, indicating potential overfitting or convergence to a local optimum through excessive iteration. Therefore, to balance model stability and efficiency, the optimal number of diffusion steps is 50, ensuring performance in anomaly detection while maximizing computational efficiency and resource utilization in practical applications.

4) *Window size*: The size of the window is also an important parameter that affects the performance of the model. For the PSM dataset, we conduct experiments on STDMD-AD using different window sizes (mainly 25, 50, 75, and 100). As shown in Fig. 8, it can be seen that as the window size increases, the anomaly detection performance of the model gradually improves. The experiment shows that window size has a significant impact on the final result and cannot be ignored. A window that is too small will reduce the amount of information in each sample, thereby affecting the model's learning of features.

E. Comparative experiment

We conduct a comparative evaluation of our STDMD-AD model against various existing models on five common publicly available anomaly detection datasets: MSL, SMAP, PSM, SMD, and SWaT. The performance metrics evaluated include Precision (P), Recall (R), and F1-Score (F1). Furthermore, we compare the STDMD-AD model with seven state-of-the-art methods based on prediction, reconstruction, and diffusion models, using evaluation metrics including Precision (P), Recall (R), and F1-Score (F1): (1) Isolation Forest (IForest) detects anomalies by isolating anomalous data points from the rest. (2) LSTM-VAE leverages Long Short-Term Memory networks and Variational Autoencoders to reconstruct data and detect sequence anomalies through reconstruction error analysis. (3) TranAD employs Transformer architectures to capture broader temporal trends in the data for anomaly prediction. (4) MTAD-GAT utilizes Graph Attention Networks to model multivariate time series (MTS) and combines prediction-based and reconstruction-based approaches to enhance representation learning. (5) GDN proposes a Graph Neural Network-based method to aggregate information among sensors for anomaly detection. (6) ImDiffusion designs a raster mask strategy to create missing values in the data and leverages diffusion models for prediction and anomaly detection. (7) DDMT integrates denoising diffusion models with Transformers for multivariate time series anomaly detection.

The selected baselines fully cover the three mainstream methodological paradigms in anomaly detection: prediction-based (TranAD, MTAD-GAT), reconstruction-based (LSTM-VAE, GDN) and diffusion-based (ImDiffusion, DDMT), enabling a systematic evaluation of our STDMD-AD model against the core research directions of the field. This setup not only highlights the superior performance of our method over traditional prediction and reconstruction-based approaches with inherent limitations elaborated in Sections 1 and 2, but also demonstrates its notable improvements over state-of-the-art emerging diffusion-based techniques. In terms of representativeness, the first five methods (IForest, LSTM-VAE, TranAD, MTAD-GAT, GDN) are well-recognized benchmark models widely adopted in top international conferences and prestigious journals, while the two diffusion-based models (ImDiffusion, DDMT) represent the latest research advances published in ICML and ICLR during 2022–2023, ensuring the representativeness and cutting-edge nature of the comparative experiments. In addition, the selected baselines span the entire performance spectrum of anomaly detection methods, ranging from classical statistical approaches (IForest) to modern deep learning architectures covering prediction-based, reconstruction-based and diffusion-based types, which offers readers a clear overview of the developmental progression of the anomaly detection field and an intuitive understanding of the relative performance positioning of our STDMD-AD model.

As shown in the results of Table III, for the MSL dataset, our model STDMD-AD achieves the highest F1 score of 94.72. In contrast, the best performing model after SDMD-AD is DDMT (F1: 94.26). For the SMAP dataset, the F1 score is 97.00,

TABLE II

THE ABLATION EXPERIMENT RESULTS OF BASELINE AND OTHER MODULES ON FIVE COMMON PUBLICLY AVAILABLE ANOMALY DETECTION DATASETS, INCLUDING MSL, SMAP, SMD, PSM, AND SWaT.

Model	MSL			SMAP			PSM			SMD			SWaT		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
STDMD-AD	91.32	98.37	94.72	99.44	94.68	97.00	99.06	98.28	98.67	88.93	80.91	84.73	95.37	93.03	94.19
Temporal mask	90.05	87.32	89.52	94.10	89.45	92.48	97.41	89.37	93.22	86.42	73.79	79.61	92.80	80.51	86.22
Spatial mask	87.13	89.46	88.28	97.95	87.57	92.47	98.64	90.75	94.53	83.30	81.72	82.50	97.84	80.63	88.41

TABLE III

COMPARATIVE EXPERIMENTAL RESULTS OF STDMD-AD AND OTHER MODELS ON FIVE COMMON PUBLICLY AVAILABLE ANOMALY DETECTION DATASETS, INCLUDING MSL, SMAP, SMD, PSM, AND SWaT.

Model	MSL			SMAP			PSM			SMD			SWaT		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
STDMD-AD	91.32	98.37	94.72	99.44	94.68	97.00	99.06	98.28	98.67	88.93	80.91	84.73	95.37	93.03	94.19
IForest [40]	50.44	55.03	52.65	54.15	58.95	55.26	49.22	56.98	52.51	41.32	77.56	52.60	45.04	79.59	57.52
LSTM-VAE [41]	77.40	93.23	84.58	81.57	45.18	58.15	97.03	39.30	55.94	75.22	85.45	80.01	40.58	83.10	54.53
TranAD [25]	89.52	99.99	93.73	86.21	56.96	68.60	96.70	73.22	83.34	85.18	80.45	82.75	90.72	79.06	84.48
MTAD-GAT [24]	85.93	93.12	89.38	47.49	99.99	64.39	90.67	88.38	89.51	75.76	84.40	79.85	82.62	80.11	81.35
ImDiffusion [36]	92.76	89.65	91.17	97.95	87.57	92.47	93.48	98.71	96.02	83.30	81.72	82.50	99.99	65.73	79.32
DDMT [42]	91.03	97.73	94.26	93.43	99.11	96.19	95.42	97.50	96.45	88.07	81.50	84.66	90.03	82.51	86.11
GDN [43]	85.84	93.12	89.33	81.04	99.99	89.52	97.59	39.09	55.82	70.85	99.78	82.85	99.99	64.25	78.24

TABLE IV

COMPARISON OF THE AVERAGE VALUES OF EVALUATION INDEXES OF STDMD-AD AND OTHER MODELS ON FIVE DATA SETS.

Model	Average		
	P	R	F1
STDMD-AD	94.82	93.05	93.86
IForest	48.03	65.62	54.10
LSTM-VAE	74.36	69.25	66.64
TranAD	89.66	77.93	82.58
MTAD-GAT	76.49	89.20	80.89
ImDiffusion	93.49	84.67	88.29
DDMT	91.59	91.67	91.53
GDN	87.06	79.24	79.15

making it the best performing dataset. The closest competitor is DDMT (F1: 96.19). For the PSM dataset, our model achieves an F1 score of 98.67, the highest among all models. The second best performing model is DDMT (F1: 96.45). For the SMD dataset, STDMD-AD achieves an F1 score of 84.73, making it one of the best performing datasets. For the SWaT dataset, our model increases the F1 score to 94.19, making it one of the best performing models. DDMT performs the best with an F1 score of 86.11. The observed performance variation across datasets can be attributed to differences in their inherent characteristics. Generally, achieving significant performance improvements becomes more challenging as the dimensionality of the data increases. For instance, PSM,

SMAP, and MSL have relatively lower dimensionalities (25-26 dimensions), while SMD contains 38 dimensions and SWaT has 51 dimensions. This explains why all models, including both our approach and the baseline methods, demonstrate relatively poorer performance on the SMD dataset compared to others. However, the superior performance of STDMD-AD on SWaT, despite its high dimensionality, may be attributed to the concentrated nature of anomalies in SWaT, as opposed to the more sparse and scattered anomaly distribution in SMD. These factors collectively contribute to the performance differences observed across the five datasets.

In Table IV, the evaluation results demonstrate the excellent performance of our STDMD-AD model across various metrics. Specifically, the STDMD-AD model achieves an average precision of 94.82, an average recall of 93.05, and an average F1 score of 93.86 across all five datasets. Compared to other models, STDMD-AD shows significantly better performance in multivariate time series anomaly detection. IForest performs poorly across all metrics, particularly in F1 score, reflecting its limitations in handling complex time series data. The LSTM-VAE model, while showing decent precision, has low recall, leading to a lower F1 score and limiting its effectiveness in anomaly detection. TranAD and MTAD-GAT outperform IForest and LSTM-VAE on some datasets but still fall short of STDMD-AD. TranAD, though close in precision, lags behind in recall and F1 score, while MTAD-GAT achieves better recall but remains weaker in precision and F1 score. The ImDiffusion model demonstrates good precision and recall, but its overall F1 score is still below that of STDMD-AD. DDMT and GDN perform relatively well but do not reach the comprehensive

capability of STDMD-AD.

In summary, our proposed STDMD-AD model, especially with the spatial-temporal masking mechanism, consistently achieves the best or near-best performance across all five datasets. This mechanism enhances robustness by mitigating the impact of anomalous data during training, thereby improving anomaly detection and increasing precision, recall, and F1 score. These results highlight the robustness and generalization capability of our approach in real-world multivariate time series anomaly detection tasks. In particular, the proposed graph attention-inspired spatiotemporal Transformer and dynamic spatial-temporal masking strategy enable precise and adaptive modeling of complex spatiotemporal dependencies between multiple sensors, which is the key to the superior performance of STDMD-AD on high-dimensional multi-sensor time series anomaly detection, and provides a new perspective for multi-sensor dependency modeling in related fields.

V. CONCLUSION

In this paper, we develop the Spatial-Temporal Dynamic Mask Diffusion for Anomaly Detection (STDMD-AD) framework, designed to enhance anomaly detection in multidimensional time series data. Our method integrates a Spatial-Temporal Transformer with a dynamic masking strategy, effectively mitigating the impact of anomalous data during training and significantly improving the model's robustness and generalization capabilities.

Comprehensive experiments conducted on five real-world datasets—SMD, PSM, MSL, SMAP, and SWaT—demonstrate that our model outperforms existing state-of-the-art methods. Specifically, the proposed STDMD-AD achieves an average precision of 94.82, a recall of 93.05, and an F1-score of 93.86 across all datasets, highlighting its effectiveness in accurately detecting anomalies in complex time series data. The success of our approach is largely attributed to the dynamic masking mechanism, which reduces the prevalence of anomalous data during training, thereby enhancing the model's ability to learn normal patterns and effectively detect deviations. Furthermore, the integration of spatial and temporal features through the Spatial-Temporal Transformer allows our model to capture intricate relationships within the data, further boosting its performance.

Despite these improvements, the iterative nature of diffusion sampling and the complexity of the Transformer architecture pose challenges for real-time deployment on resource-constrained IIoT edge devices. Consequently, future work will focus on three strategic directions to bridge this gap. First, we will investigate model compression techniques, including knowledge distillation and network quantization, and explore Latent Diffusion Models (LDMs) to reduce computational complexity while preserving representation capability. Second, to facilitate scalable and secure deployment in distributed industrial networks, we aim to extend STDMD-AD to a Federated Learning (FL) paradigm. Drawing inspiration from efficient aggregation protocols like LSFL [44], we plan to incorporate lightweight secure aggregation to ensure robustness against bandwidth constraints and Byzantine attacks. Finally,

we will develop adaptive mechanisms that allow the model to dynamically adjust masking ratios and diffusion steps based on real-time noise levels and device resource availability.

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